

3.2 Let $x(n)$ be a stationary random process with zero mean and autocorrelation $r_x(k)$. We form the process, $y(n)$, as follows

$$y(n) = x(n) + f(n)$$

where $f(n)$ is a known deterministic sequence. Find the mean $m_y(n)$ and the autocorrelation $r_y(k, l)$ of the process $y(n)$.

Solution

The mean of the process is

$$m_y(n) = E\{y(n)\} = E\{x(n)\} + f(n) = f(n)$$

and the autocorrelation is

$$\begin{aligned} r_y(k, l) &= E\{y(k)y(l)\} = E\{[x(k) + f(k)][x(l) + f(l)]\} \\ &= E\{x(k)x(l)\} + f(k)f(l) = r_x(k, l) + f(k)f(l) \end{aligned}$$

3.8 Consider the random process

$$x(n) = A \cos(n\omega + \phi) + w(n)$$

where $w(n)$ is zero mean white Gaussian noise with a variance σ_w^2 . For each of the following cases, find the autocorrelation sequence and if the process is WSS, find the power spectrum.

- (a) A is a Gaussian random variable with zero mean and variance σ_A^2 and both ω and ϕ are constants.
- (b) ϕ is uniformly distributed over the interval $[-\pi, \pi]$ and both A and ω are constants.
- (c) ω is a random variable that is uniformly distributed over the interval $[\omega_0 - \Delta, \omega_0 + \Delta]$ and both A and ϕ are constants.

Solution

- (a) When ω and ϕ are constants, then

$$r_x(k, l) = E\{x(k)x(l)\} = E\{A \cos(k\omega + \phi)A \cos(l\omega + \phi)\} + \sigma_w^2 \delta(k - l)$$

Thus,

$$\begin{aligned} r_x(k, l) &= E\{A^2\} \cos(k\omega + \phi) \cos(l\omega + \phi) + \sigma_w^2 \delta(k - l) \\ &= \sigma_A^2 \cos(k\omega + \phi) \cos(l\omega + \phi) + \sigma_w^2 \delta(k - l) \end{aligned}$$

Note that since $r_x(k, l)$ does not depend on the difference $k - l$, then $x(n)$ is not wide-sense stationary, and the power spectrum is not defined for this process.

- (b) When A and ω are constants and ϕ is a random variable that is uniformly distributed over the interval $[-\pi, \pi]$, then the autocorrelation is

$$\begin{aligned} r_x(k, l) &= E\{A^2 \cos(k\omega + \phi) \cos(l\omega + \phi)\} + \sigma_w^2 \delta(k - l) \\ &= \frac{1}{2} A^2 E\{\cos[(k + l)\omega + 2\phi]\} + \frac{1}{2} A^2 E\{\cos(k - l)\omega\} + \sigma_w^2 \delta(k - l) \end{aligned}$$

However, since $E\{\cos[(k + l)\omega + 2\phi]\} = 0$, then the autocorrelation is

$$r_x(k, l) = \frac{1}{2} A^2 \cos(k - l)\omega + \sigma_w^2 \delta(k - l)$$

Therefore, $r_x(k, l)$ depends on the difference $(k - l)$, and the process is WSS. The power spectrum is

$$P_x(e^{j\omega}) = \frac{\pi A^2}{2} \delta(\omega - \omega) + \frac{\pi A^2}{2} \delta(\omega + \omega) + \sigma_w^2$$

- (c) As in parts (a) and (b), the autocorrelation of the process $x(n)$ is

$$r_x(k, l) = E\{x(k)x(l)\} = E\{A \cos(k\omega + \phi) A \cos(l\omega + \phi)\} + \sigma_w^2 \delta(k - l)$$

In this case, however, ω is a random variable, and the expectation of the product of the cosines is

$$E\{A \cos(k\omega + \phi) A \cos(l\omega + \phi)\} = A^2 E\left\{\frac{1}{2} \cos[(k - l)\omega] + \frac{1}{2} \cos[(k + l)\omega + 2\phi]\right\}$$

Since ω is uniformly distributed over the interval $[\omega_0 - \Delta, \omega_0 + \Delta]$, the expectation of the first term is

$$\begin{aligned} E\{\cos[(k - l)\omega]\} &= \frac{1}{2\Delta} \int_{\omega_0 - \Delta}^{\omega_0 + \Delta} \cos[(k - l)\omega] d\omega \\ &= \frac{1}{2\Delta(k - l)} \left[\sin[(k - l)(\omega_0 + \Delta)] - \sin[(k - l)(\omega_0 - \Delta)] \right] \end{aligned}$$

With ϕ a constant, the expectation of the second term is

$$E\{\cos[(k + l)\omega + 2\phi]\} = \frac{1}{2\Delta(k + l)} \left[\sin[(k + l)(\omega_0 + \Delta + 2\phi)] - \sin[(k + l)(\omega_0 - \Delta + 2\phi)] \right]$$

Therefore, $x(n)$ is *not* WSS. However, if ϕ is a random variable that is uniformly distributed over the interval $[-\pi, \pi]$, then this second expectation is zero, and the autocorrelation becomes

$$\begin{aligned} r_x(k, l) &= \frac{A^2}{4\Delta(k - l)} \left\{ \sin[(k - l)(\omega_0 + \Delta)] - \sin[(k - l)(\omega_0 - \Delta)] \right\} \\ &= \frac{A^2}{2\Delta(k - l)} \sin[(k - l)\Delta] \cos[(k - l)\omega_0] \end{aligned}$$

and the process is WSS. With an autocorrelation sequence given by

$$r_x(k) = \frac{\pi A^2}{2\Delta} \frac{\sin k\Delta}{\pi k} \cos k\omega_0$$

using the DTFT pair

$$r_x(k) = \frac{\sin k\Delta}{\pi k} \iff P_x(e^{j\omega}) = \begin{cases} 1 & ; \quad |\omega| \leq \Delta \\ 0 & ; \quad \text{else} \end{cases}$$

it follows that the power spectrum of $x(n)$ is

$$P_x(e^{j\omega}) = \begin{cases} \frac{\pi A^2}{4\Delta} & ; \quad \omega_0 - \Delta \leq |\omega| \leq \omega_0 + \Delta \\ 0 & ; \quad \text{else} \end{cases}$$

3.14 For each of the following, determine whether or not the random process is

1. Wide-sense stationary.
 2. Mean ergodic.
- (a) $x(n) = A$ where A is a random variable with probability density function $f_A(\alpha)$.
 - (b) $x(n) = A \cos(n\omega_0)$ where A is a Gaussian random variable with mean m_A and variance σ_A^2 .
 - (c) $x(n) = A \cos(n\omega_0 + \phi)$ where ϕ is a random variable that is uniformly distributed between $-\pi$ and π .
 - (d) $x(n) = A \cos(n\omega_0) + B \sin(n\omega_0)$ where A and B are uncorrelated zero mean random variables with variance σ^2 .
 - (e) A Bernoulli process with $\Pr\{x(n) = 1\} = p$ and $\Pr\{x(n) = -1\} = 1 - p$.
 - (f) $y(n) = x(n) - x(n-1)$ where $x(n)$ is the Bernoulli process defined in part (e).

Solution

- (a) We are given a process $x(n) = A$, where A is a random variable with probability density function $f_A(\alpha)$. To check for wide-sense stationarity we need to compute the mean and autocorrelation of $x(n)$. The mean of this process is

$$m_x(n) = E\{x(n)\} = E\{A\}$$

which is a constant. The autocorrelation is

$$r_x(k, l) = E\{x(k)x(l)\} = E\{A^2\}$$

which is also a constant. Therefore, $x(n)$ is WSS.

To check for ergodicity in the mean, note that

$$c_x(k) = r_x(k) - m_x^2 = E\{A^2\} - E^2\{A\}$$

which is a constant. Therefore, $x(n)$ is ergodic in the mean only if the variance of A is zero, $c_x(k) = 0$.

- (b) With $x(n) = A \cos(n\omega_0)$, note that the mean of the process is

$$m_x(n) = E\{x(n)\} = E\{A \cos(n\omega_0)\} = E\{A\} \cos(n\omega_0) = m_A \cos(n\omega_0)$$

which depends on n . Thus, $x(n)$ is not WSS and, therefore, not ergodic in the mean.

- (c) For $x(n) = A \cos(n\omega_0 + \phi)$ with ϕ a random variable that is uniformly distributed between $-\pi$ and π , the mean of $x(n)$ is

$$m_x(n) = E\{x(n)\} = E\{A \cos(n\omega_0 + \phi)\} = \frac{A}{2\pi} \int_{-\pi}^{\pi} \cos(n\omega_0 + \phi) d\phi = 0$$

which is a constant. For the autocorrelation we have

$$\begin{aligned} r_x(k, l) &= E\{A \cos(k\omega_0 + \phi) A \cos(l\omega_0 + \phi)\} \\ &= \frac{1}{2} A^2 E\{\cos([k-l]\omega_0) + \cos([k+l]\omega_0 + 2\phi)\} \\ &= \frac{1}{2} A^2 \cos(k-l)\omega_0 \end{aligned}$$

which is a function of $(k-l)$. Therefore, $x(n)$ is WSS.

To check for ergodicity in the mean, note that

$$\begin{aligned} \frac{1}{N} \sum_{k=0}^{N-1} c_x(k) &= \frac{1}{2} \frac{A^2}{N} \sum_{k=0}^{N-1} \cos k\omega_0 \\ &= \frac{1}{2} \frac{A^2}{N} \sum_{k=0}^{N-1} \left\{ \frac{1}{2} e^{jk\omega_0} + \frac{1}{2} e^{-jk\omega_0} \right\} \\ &= \frac{1}{4} \frac{A^2}{N} \left\{ \frac{1 - e^{jN\omega_0}}{1 - e^{j\omega_0}} + \frac{1 - e^{-jN\omega_0}}{1 - e^{-j\omega_0}} \right\} \\ &= \frac{A^2 \sin(N\omega_0/2)}{2N \sin(\omega_0/2)} \cos\left(\frac{N-1}{2}\omega_0\right) \end{aligned}$$

which goes to zero as $N \rightarrow \infty$, provided that $\omega_0 \neq 0$. If $\omega_0 = 0$, then $x(n) = A \cos \phi$ and $c_x(k) = A^2/2$. In this case, $x(n)$ is not ergodic in the mean. Therefore, $x(n)$ is ergodic in the mean only if $\omega_0 \neq 0$.

- (d) The mean of this process is

$$E\{x(n)\} = E\{A\} \cos(n\omega_0) + E\{B\} \sin(n\omega_0)$$

Since $E\{A\} = E\{B\} = 0$, then $E\{x(n)\} = 0$. For the autocorrelation we have

$$\begin{aligned} r_x(k, l) &= E\{x(k)x(l)\} = E\{[A \cos(\omega_0 k) + B \sin(\omega_0 k)][A \cos(\omega_0 l) + B \sin(\omega_0 l)]\} \\ &= E\{A^2\} \cos(k\omega_0) \cos(l\omega_0) + E\{B^2\} \sin(k\omega_0) \sin(l\omega_0) \\ &\quad + E\{AB\} [\cos(k\omega_0) \sin(l\omega_0) + \sin(k\omega_0) \cos(l\omega_0)] \end{aligned}$$

Since A and B are uncorrelated and have zero mean, then $E\{AB\} = 0$ and $E\{A^2\} = E\{B^2\} = \sigma^2$. Therefore, we have

$$r_x(k, l) = \sigma^2 [\cos(k\omega_0) \cos(l\omega_0) + \sin(k\omega_0) \sin(l\omega_0)] = \sigma^2 \cos(k-l)\omega_0$$

Since the mean is a constant and the correlation function $r_x(k, l)$ depends only the difference, $k-l$, then $x(n)$ is a wide-sense stationary process.

As in part (c), $x(n)$ is ergodic in the mean only if $\omega_0 \neq 0$.

- (e) With $x(n)$ a Bernoulli process with $\Pr\{x(n) = 1\} = p$ and $\Pr\{x(n) = -1\} = 1 - p$, the mean of $x(n)$ is

$$m_x(n) = E\{x(n)\} = p - (1 - p) = 2p - 1$$

and the autocorrelation is

$$r_x(k, l) = E\{x(k)x(l)\} = \begin{cases} E\{x^2(k)\} & ; \quad k = l \\ E\{x(k)x(l)\} & ; \quad k \neq l \end{cases}$$

Therefore,

$$r_x(k, l) = \begin{cases} p + (1-p) = 1 & ; \quad k = l \\ (1-2p)^2 & ; \quad k \neq l \end{cases}$$

or,

$$r_x(k, l) = 4p(1-p)\delta(k-l) + (1-2p)^2$$

which is a function of $k-l$. Therefore, $x(n)$ is WSS. With

$$c_x(k) = r_x(k) - m_x^2 = 4p(1-p)\delta(k)$$

it follows that

$$\frac{1}{N} \sum_{k=0}^{N-1} c_x(k) = \frac{1}{N} 4p(1-p) \rightarrow 0 \text{ as } N \rightarrow \infty$$

Thus, $x(n)$ is ergodic in the mean.

- (f) With $y(n) = x(n) - x(n-1)$ where $x(n)$ is a Bernoulli process, wide-sense stationarity may be easily checked using the direct approach taken in parts (a)-(e) of this problem. However, it is easier to note that since $y(n)$ is the response of a linear shift-invariant system to an input that is a WSS process, then $y(n)$ will be WSS.

For ergodicity in the mean, note that since

$$c_x(k) = 4p(1-p)\delta(k)$$

then

$$c_y(k) = c_x(k) * h(k) * h(-k) = 4p(1-p)[2\delta(k) - \delta(k-1) - \delta(k+1)]$$

and, clearly,

$$\frac{1}{N} \sum_{k=0}^{N-1} c_x(k) \rightarrow 0 \text{ as } N \rightarrow \infty$$

- 3.17** Given a wide-sense stationary random process $x(n)$, we would like to design a “linear predictor” that will predict the value of $x(n+1)$ using a linear combination of $x(n)$ and $x(n-1)$. Thus, our predictor for $x(n+1)$ is of the form

$$\hat{x}(n+1) = ax(n) + bx(n-1)$$

where a and b are constants. Assume that the process has zero mean

$$E\{x(n)\} = 0$$

and that we want to minimize the mean-square error

$$\xi = E\{[x(n+1) - \hat{x}(n+1)]^2\}$$

- With $r_x(k)$ the autocorrelation of $x(n)$, determine the optimum predictor of $x(n)$ by finding the values of a and b that minimize the mean-square error.
- What is the minimum mean-square error of the predictor? Express your answer in terms of the autocorrelation $r_x(k)$.
- If $x(n+1)$ is uncorrelated with $x(n)$, what form does your predictor take?
- If $x(n+1)$ is uncorrelated with both $x(n)$ and $x(n-1)$, what form does your predictor take?

Solution

- The mean-square error that we want to minimize is

$$\xi = E\{[x(n+1) - \hat{x}(n+1)]^2\} = E\{x^2(n+1) - 2x(n+1)\hat{x}(n+1) + \hat{x}^2(n+1)\}$$

Since the estimate of $x(n+1)$ is

$$\hat{x}(n+1) = ax(n) + bx(n-1)$$

then setting the derivative of ξ with respect to a and b equal to zero we have

$$\begin{aligned}\frac{\partial \xi}{\partial a} &= -2E\{x(n+1)x(n)\} + E\{2\hat{x}(n+1)x(n)\} = 0 \\ \frac{\partial \xi}{\partial b} &= -2E\{x(n+1)x(n-1)\} + E\{2\hat{x}(n+1)x(n-1)\} = 0\end{aligned}$$

Dividing by 2 and substituting for $\hat{x}(n+1)$ gives

$$\begin{aligned}aE\{x^2(n)\} + bE\{x(n)x(n-1)\} &= E\{x(n+1)x(n)\} \\ aE\{x(n)x(n-1)\} + bE\{x^2(n-1)\} &= E\{x(n+1)x(n-1)\}\end{aligned}$$

Putting these equations in matrix form we have

$$\begin{bmatrix} r_x(0) & r_x(1) \\ r_x(1) & r_x(0) \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} r_x(1) \\ r_x(2) \end{bmatrix}$$

Solving for a and b we find

$$\begin{bmatrix} a \\ b \end{bmatrix} = \frac{1}{r_x^2(0) - r_x^2(1)} \begin{bmatrix} r_x(0)r_x(1) - r_x(1)r_x(2) \\ r_x(0)r_x(2) - r_x^2(1) \end{bmatrix}$$

(b) For the minimum mean-square error we have

$$\begin{aligned}\xi &= E\left\{[x(n+1) - \hat{x}(n+1)]^2\right\} = E\{[x(n+1) - \hat{x}(n+1)][x(n+1) - \hat{x}(n+1)]\} \\ &= E\{[x(n+1) - \hat{x}(n+1)]x(n+1)\} - E\{[x(n+1) - \hat{x}(n+1)]\hat{x}(n+1)\}\end{aligned}$$

Note that for values of a and b that minimize the mean-square error, the derivatives of ξ with respect to a and b are equal to zero, which implies that the second term in the equation above is equal to zero. Therefore, the minimum mean-square error is

$$\xi = E\{[x(n+1) - \hat{x}(n+1)]x(n+1)\} = E\{[x(n+1) - ax(n) - bx(n-1)]x(n+1)\}$$

or,

$$\begin{aligned}\xi &= r_x(0) - ar_x(1) - br_x(2) \\ &= r_x(0) - r_x(1)\frac{r_x(0)r_x(1) - r_x(1)r_x(2)}{r_x^2(0) - r_x^2(1)} - r_x(2)\frac{r_x(0)r_x(2) - r_x^2(1)}{r_x^2(0) - r_x^2(1)}\end{aligned}$$

(c) If $x(n+1)$ and $x(n)$ are uncorrelated, then $r_x(1) = 0$, and the values for a and b become

$$a = 0 \quad ; \quad b = r_x(2)/r_x(0)$$

In this case, the linear predictor is

$$\hat{x}(n+1) = \frac{r_x(2)}{r_x(0)}x(n-1)$$

(d) If $x(n+1)$ is uncorrelated with both $x(n)$ and $x(n-1)$, then the values for a and b are

$$a = b = 0$$

and the linear predictor is

$$\hat{x}(n+1) = 0$$

which is equal to the expected value of $x(n+1)$,

$$\hat{x}(n+1) = E\{x(n+1)\}$$
